LASER VAULT HEIST

C++17 / OpenGL 3.3 core

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Topic 12



The heist and its constraints

6 laser barriers

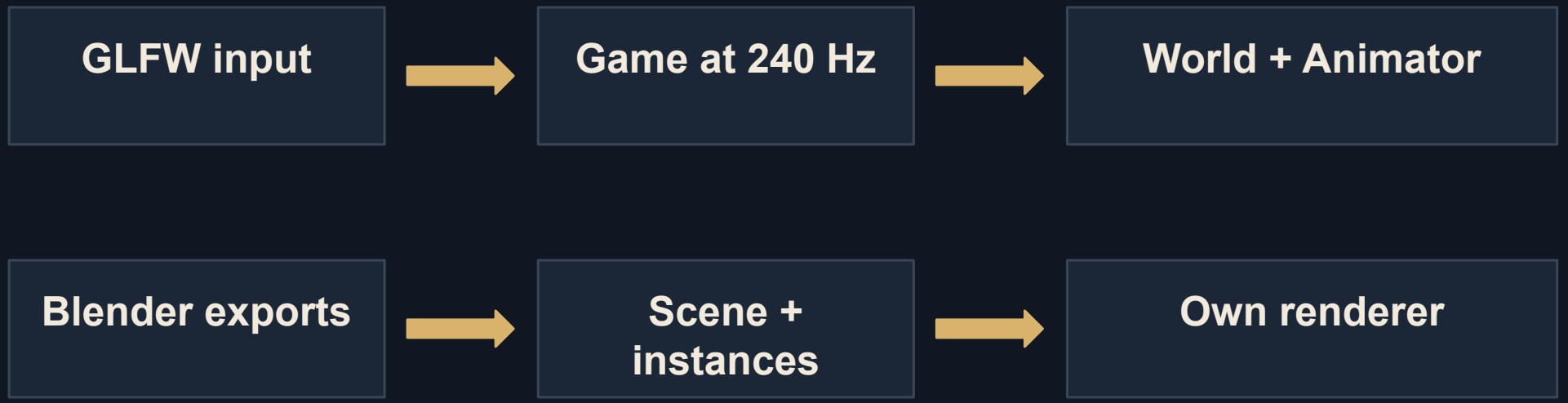
Walk, wait, crouch
and reach the safe.

Lamp / CCTV / safe
Three meaningful interactions

Failure -> retry -> victory



Architecture and frame pipeline

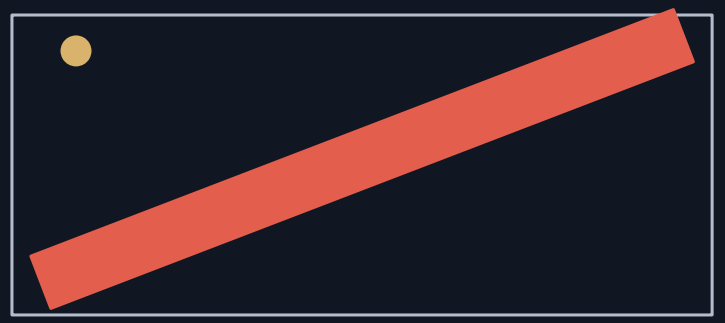


Retained input edges; pause stops simulation



Collision in the laser’s local space

q = inverse(M) * vec4(p, 1)



World AABB can be too large

Beam-local box



abs(q.x) <= half.x
abs(q.y) <= half.y
abs(q.z) <= half.z

Same model matrix for the visible beam and contact test

Per-fragment lighting



H = normalize(L + V)

Diffuse: max(N·L, 0)
Specular: max(N·H, 0)^s

Attenuation + soft cones
Inverse-transpose normals
One PCF shadow map

Original character and animation





20 bones

21,782 triangles

5 clips
0.2 s blends
Lerp + slerp

$$p' = \Sigma\, w_i\, G_i(t)\, B_i^{-1}\, p$$

Live CCTV with render-to-texture

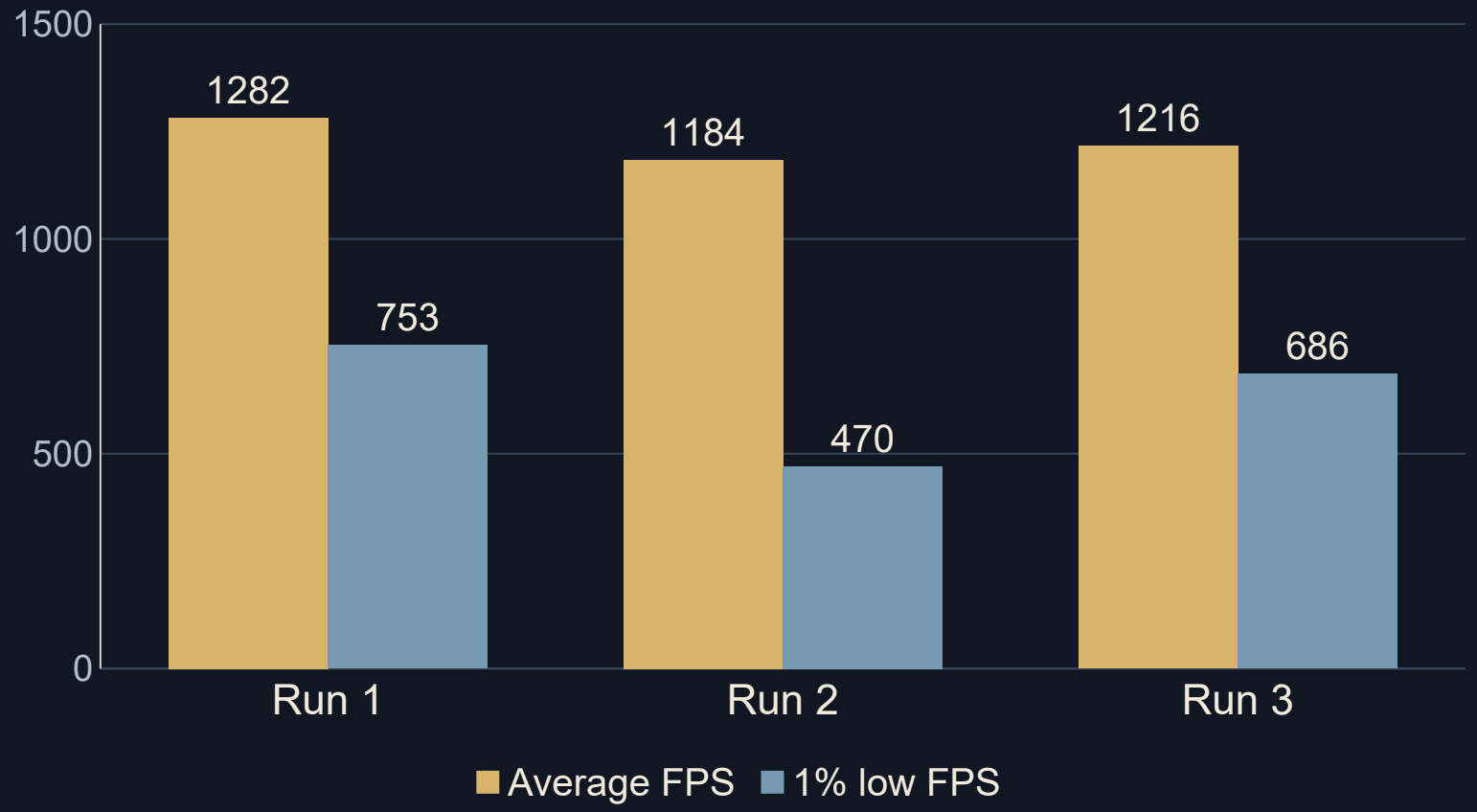
640 × 360 FBO

Color + depth
Two live cameras
Animated thief

Exclude the screen
from its own pass.



Measured local performance



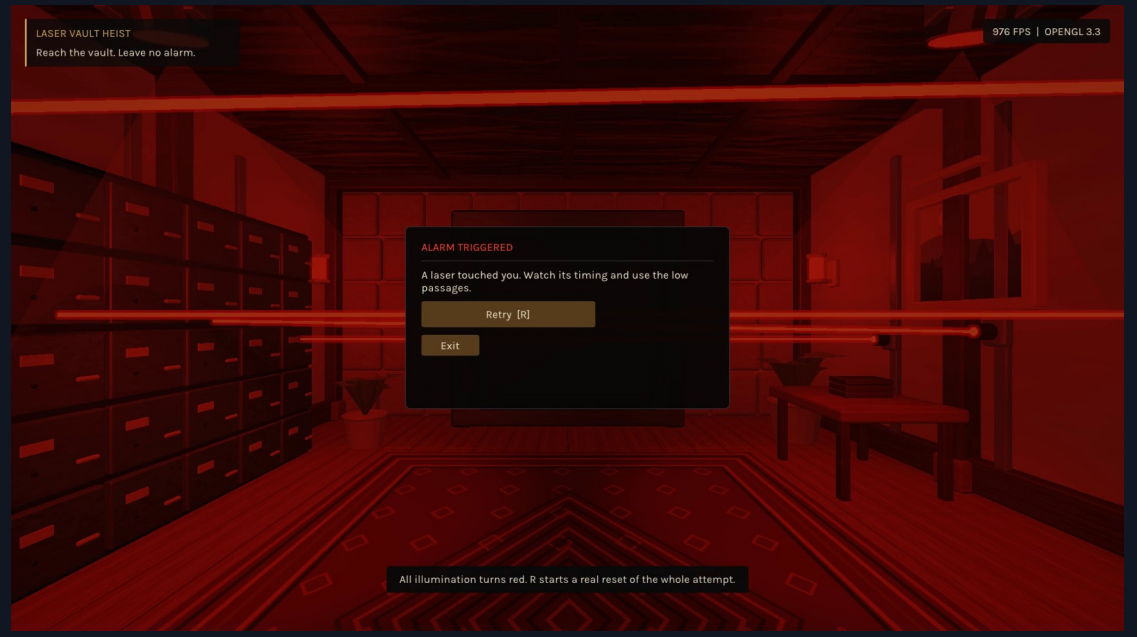
1280 × 720

i5-13400F
RTX 5060
32 GiB RAM

3 × 60 s measured
5 s warm-up
VSync off

TA hardware remains untested. Raw frame times are included.

Failure, reset and safe-opening victory



Alarm -> R -> fresh attempt



Wheel -> bolts -> hinged door

The final low-beam passage and opening are continuous in the demo.

Results, limits and next checks

Complete local implementation

Original assets and five clips
Three passing test suites
18 verified research references
3:30 real-time demo



Next checks

Human input • TA machine • live defense